



9. Exploratory Data Analysis

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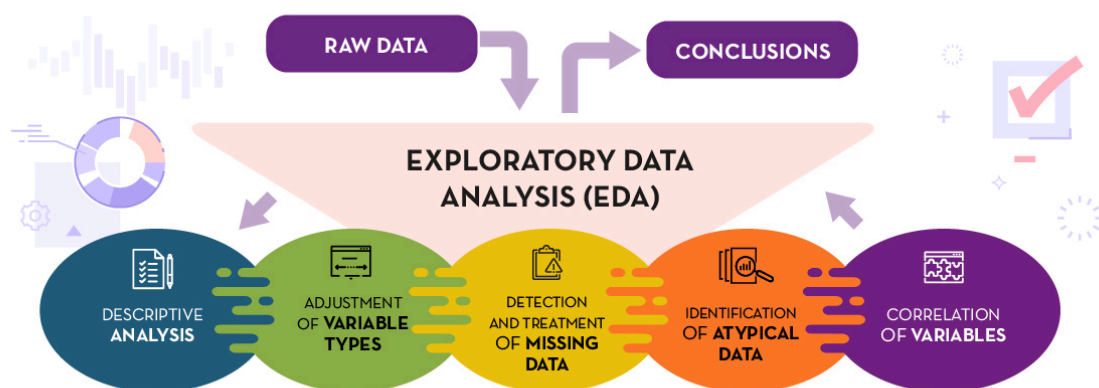
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Exploratory Data Analysis (EDA) is one of the first and most important steps in the data analytics workflow. Before applying statistical models or machine

learning algorithms, we need to understand the structure, quality, and characteristics of the dataset. EDA combines descriptive statistics and data visualization to uncover patterns, detect anomalies, and generate insights that guide subsequent analysis.

1. What is Exploratory Data Analysis (EDA)?

Exploratory Data Analysis (EDA) is the process of investigating and summarizing data using statistical methods and visualizations. The objective is to understand the underlying structure of the data, discover patterns, identify unusual observations, and explore relationships between variables before performing more advanced analysis.

Common tools used in EDA include summary statistics, tables, histograms, box plots, scatter plots, and correlation analysis.

2. Why is EDA Important?

EDA provides valuable insights that improve the quality and reliability of data analysis. It helps analysts understand their data before making assumptions or building predictive models.

Some important benefits of EDA include:

- Detecting missing values, outliers, and anomalies.
- Identifying hidden trends and data distributions.
- Validating assumptions required for statistical analysis.
- Understanding relationships between variables.
- Supporting feature selection and feature engineering.
- Improving the performance and interpretability of machine learning models.

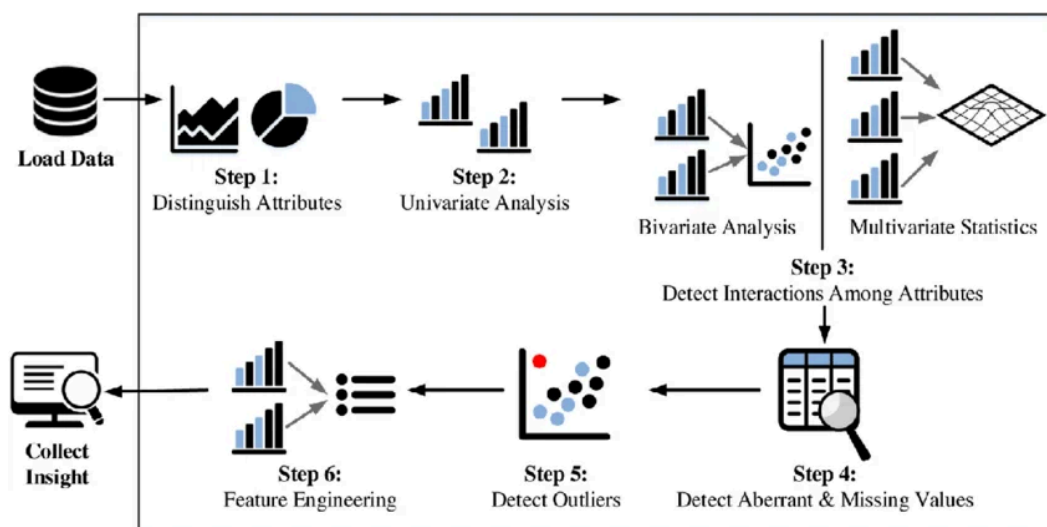
3. The EDA Process

Although the exact workflow depends on the dataset and application, a typical exploratory data analysis process consists of the following steps:

- **Descriptive Analysis:** Summarize the dataset using statistical measures and visualizations to understand central tendency, variability, and data distribution.
- **Adjustment of Variable Types:** Verify that each feature has the correct data type (categorical, numerical, datetime, etc.) before analysis.
- **Missing Data Handling:** Detect missing values and apply appropriate strategies such as removing records or imputing missing values.
- **Outlier Detection:** Identify unusual observations that may indicate data errors or significantly influence statistical analysis and machine learning models.
- **Correlation Analysis:** Examine relationships between variables to identify dependencies, redundancy, and potential multicollinearity.

4. Types of Exploratory Data Analysis

EDA can be categorized according to the number of variables being analyzed.



- **Univariate Analysis:** Univariate analysis focuses on a **single variable**. The objective is to understand its distribution and summarize its characteristics using descriptive statistics.

Common techniques include:

- Mean, Median, and Mode

- Variance and Standard Deviation
- Quartiles
- Histograms
- Box Plots
- **Bivariate Analysis:** Bivariate analysis studies the relationship between **two variables**. It helps determine whether changes in one variable are associated with changes in another.

Common techniques include:

- Covariance
- Correlation
- Scatter Plots
- Regression Lines
- **Multivariate Analysis:** Multivariate analysis examines **three or more variables simultaneously** to uncover more complex relationships that cannot be observed using univariate or bivariate methods.

Common techniques include:

- Pair Plots
- Correlation Matrices
- Time Series Analysis
- Spatial Analysis
- Dimensionality Reduction techniques (e.g., PCA)

Summary

Exploratory Data Analysis is an essential step that bridges raw data and meaningful insights. By combining descriptive statistics with visualization techniques, EDA helps analysts understand data quality, identify patterns and relationships, detect outliers, and prepare datasets for statistical analysis and machine learning.

Throughout this chapter, we will use **Pandas** together with visualization libraries to perform these common EDA tasks efficiently.

References

<https://www.geeksforgeeks.org/data-analysis/what-is-exploratory-data-analysis/>

<https://www.ibm.com/think/topics/exploratory-data-analysis>

Next Section:

 [9.1. Statistical Methods in Data](#)